

(IAM) - IDENTITY AND ACCESS MANAGEMENT (IAM)

❖ Introduction to IAM Services

- AWS Identity and Access Management (IAM) is a web service that enables you to securely manage access to AWS services and resources.
- It allows you to control who is authenticated (signed in) and authorized (has permissions) to use resources.
- IAM provides the infrastructure necessary to control authentication and authorization for your AWS account.

➤ Key Features of IAM

▪ Centralized Access Control

- Manage who can access AWS resources **from a single place**.

▪ Granular Permissions

- Assign **fine-grained permissions** using policies (e.g., allow only read access to S3).

▪ User and Group Management

- Create **individual users** and group them to apply common permissions.

▪ Role-Based Access

- Create **roles** for services or external users (like EC2, Lambda, or third-party apps).

▪ Temporary Security Credentials

- IAM can provide **temporary access** (STS) for short-term tasks—improves security.

▪ Policy Types

- Supports **inline and managed policies** (AWS managed or custom).

▪ How IAM Works:

✓ *IAM operates by authenticating entities (such as users or applications) and authorizing their access to AWS resources based on defined policies.*

▪ *The process involves:*

1. Authentication

- IAM users **log in** with a username & password (or access keys for CLI/API).
- For extra security, **MFA (Multi-Factor Authentication)** can be enabled.

2. Authorization

- Once a user is **authenticated**, **IAM** checks the attached policies to decide whether the requested action is **allowed** or **denied**.

2. Identity Creation

- You create **IAM Users, Groups, or Roles** in your AWS account.

3. Permissions Assignment

- You attach **policies** (written in JSON) that define **what actions are allowed or denied** on specific AWS resources.

4. Temporary Access (Roles & STS)

- IAM **Roles** are used for **temporary access**, especially for services (like EC2) or external users (like AWS accounts, apps).
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❖ Getting Started with IAM

- To begin using IAM, you can create users and groups, assign permissions through policies, and manage security credentials.
- IAM integrates with AWS services, allowing you to control access to resources across your AWS environment.

❖ What is a Group in IAM?

- An IAM Group is a collection of IAM users that you manage as a single unit.

❖ Core Components of IAM

Component	Description
Users	Represents a single person or application with access credentials.
Groups	A collection of users managed as a unit; permissions assigned to a group are inherited by its users.
Roles	Used to grant temporary access to AWS services; assumed by trusted entities like applications or AWS services.
Policies	IAM Policies are JSON documents that define permissions—what actions are allowed or denied on specific AWS resources

❖ Access Types in AWS IAM

- IAM supports two main types of access to AWS resources:
 - **Programmatic Access**
 - Used to access AWS **through CLI, SDKs, or APIs**.
 - IAM users are given **Access Key ID** and **Secret Access Key**.
 - Example: A developer writes a script using AWS SDK to upload files to S3.
 - **Console Access**
 - Used **to log in** to the AWS Management **Console** (web-based UI).
 - IAM users are given a **username** and **password**.
 - You can also enable MFA (Multi-Factor Authentication) for more security.